Bayesian Reservation-Value Matching: Sequential Reciprocal Introductions under Uncertainty

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The Sequential Matching Problem: Applied Machine Learning Programme & Hackathon by Romeo & Juliet Ltd. × IIT Madras. Round 1 research, 9 October 2026

Abstract

We study the sequential reciprocal matching problem posed by The Sequential Matching Problem (Applied Machine Learning Programme & Hackathon by Romeo & Juliet Ltd. × IIT Madras): each simulated day, a policy must decide which clarifying questions to ask under a fixed budget and which feasible, non-overlapping introductions to propose, in a changing pool with incomplete observations and delayed, selective feedback. We propose Bayesian Reservation-Value Matching (BRVM). BRVM factorises the target outcome, Mutual Second-Meeting Intention (MSMI), into observable stages modelled by a hierarchical Bayesian model with person- and episode-level effects, correlated reciprocal acceptance and explicit censoring. Allocation solves a global maximum-weight matching in which each edge carries a Thompson-sampled outcome probability minus the reservation values of both endpoints, estimated as linear-programming dual prices over sampled future arrivals. This lets the policy decide when waiting is worth more than introducing. Clarification is chosen by time-aware value of information, and a scenario-aware meta-policy adapts hyperparameters to the observed world. All components run inside a validity shell that guarantees a valid response within the 10-second limit. We state seven testable hypotheses, an ablation plan against the supplied greedy, no-clarification and random-feasible baselines, and the failure cases we expect. No empirical results are claimed in this work.

1 Introduction

We propose **Bayesian Reservation-Value Matching (BRVM)**, a decision policy that chooses which clarifying questions to ask and which non-overlapping introductions to propose each simulated day. It combines four ideas, each aimed at a specific difficulty in the challenge:

- **A hierarchical Bayesian outcome model** that factorises Mutual Second-Meeting Intention (MSMI) into its observable stages (directional responses, date, mutual second-meeting Yes), with person-level and episode-level effects. It handles cold start through priors, missingness through explicit status features, and delayed outcomes through censoring.
- **Global max-weight matching on opportunity-adjusted edge values.** Each pair's value is its posterior MSMI probability minus the *reservation values* of both people, i.e. what each could be expected to achieve by waiting. Reservation values are estimated as LP dual prices over sampled future arrivals, so the policy knows when to wait and when to act.
- **Thompson sampling over feasible edges only**, giving principled exploration while hard constraints remain untouchable.
- **Value-of-information (VOI) clarification**, spending the 12-unit daily budget where an unresolved answer blocks the most expected value, weighted by how long that person remains useful.

These are wrapped in a **validity shell** that re-checks every batch, catches every exception and enforces a time budget, because a single invalid episode out of 120 makes a submission ineligible. Hyperparameters are tuned directly on the ranking objective (equal-weighted mean of the six scenario-family means), and a

lightweight **scenario-aware meta-policy** adapts settings to the observed world using only observable signals.

> ***Central hypothesis.*** *Opportunity-adjusted global allocation with Bayesian uncertainty and VOI-driven clarification will outperform the supplied greedy, no-clarification and random-feasible baselines on MSMI per 100 arrived members, with the largest gains in the cold-start, sparse-geography and shifted-condition families. All effects are stated as hypotheses; no results are claimed in this work.*

2 Problem framing: what the score rewards

The primary score is MSMI per 100 arrived members, `100 × M ÷ N`, averaged within each of six scenario families (20 private seeds each) and then across families with equal weight. Several design consequences follow directly from the rules.

Rule in the specification	Consequence for policy design
Any invalid action, malformed output, runtime failure or timeout invalidates the episode, and one invalid episode makes the entry ineligible.	Robustness is the first objective. A defensive shell (Section 5) has priority over every modelling feature.
The score counts qualifying pairs; failed introductions carry no direct penalty.	Volume matters, but every introduction locks both people for up to 7 days and permanently burns that pair. The real cost of a weak introduction is opportunity cost, which we model explicitly (Section 8).
Six families are equally weighted; best-seed selection is not allowed.	A policy tuned only to the standard family is fragile. We tune per family and adapt online (Section 10).
Decision horizon of 60 days, followed by 40 days of follow-up.	An introduction on day 59 can still qualify (date within 30 days, then Yes within 3 days). Late in the horizon, waiting has no value; the policy should become aggressive.
Denominator N includes everyone who arrived by day 59, including people who left or were never served.	Leaving hard-to-match people unserved lowers coverage, the first tie-breaker. Scarcity-aware weighting helps here.
Tie-breakers: coverage, mutual acceptances, lower clarification cost, lower inference time.	Do not spend clarification budget with zero expected value; keep inference fast.
Probability estimates are optional and do not affect ranking.	Bayesian modelling must earn its place through better decisions (exploration, waiting, clarification), not through reported probabilities.

2.1 Why pair ranking alone fails

Section 2 of the specification shows that greedily taking the best pair (A–B, 0.90) leaves C–D (0.05) for a total of 0.95, while A–D plus B–C yields 1.30. Reciprocal recommenders that score each direction of a pair [9] address only part of this: our allocation solves the matching jointly. The sequential setting adds a second failure mode: greedy matching today ignores who might arrive tomorrow and how long each person will be locked. BRVM addresses both: global matching for the first, reservation values for the second.

3 Hypotheses

Each hypothesis is paired with the ablation that tests it (Section 12).

ID	Hypothesis	Test
H1	Global max-weight matching yields more MSMI than greedy pair ranking under identical edge values.	A1: replace Blossom matching with greedy

ID	Hypothesis	Test
H2	Subtracting reservation values (waiting) improves MSMI over always matching every positive edge, most strongly in dense pools and early in the horizon.	A2: $\lambda = 0$; A3: heuristic λ vs LP-dual λ
H3	Thompson sampling improves MSMI over posterior-mean matching in cold-start and shifted families by learning person-level responsiveness faster.	A4: posterior mean instead of sample
H4	VOI-based clarification yields more MSMI per clarification unit than random or no clarification.	A5: random asks; A6: no asks
H5	Online episode-level offsets and the meta-policy reduce degradation under shifted outcome weights and changing response conditions.	A7: no online updates; A9: single parameter set
H6	Modelling correlated reciprocal acceptance improves ranking of pairs relative to an independence assumption.	A8: independent directional model
H7	A tapered scarcity bonus increases coverage with negligible MSMI loss.	A11: no scarcity bonus

4 System overview

BRVM is organised as separate layers so that normalisation, modelling and allocation stay independent, as Section 15 of the specification recommends for later Vouchsafe integration.

Layer	Role	Key method
0. Validity shell	Guarantee that every response is valid and on time	Local batch re-validation, exception fallback, time guard, memory cap
1. Feasibility	Classify every pair as feasible, infeasible or blocked by missing hard information	Supplied eligibility function; never overridden
2. Outcome model	Estimate P(MSMI) per feasible pair with uncertainty	Hierarchical Bayesian staged model, Laplace posterior, conjugate online updates
3. Allocation	Choose a non-overlapping batch, or nobody	Thompson-sampled edge values minus LP-dual reservation values; Blossom matching
4. Clarification	Choose asks within 12 units	Time-aware value of information; exact knapsack
5. Meta-policy	Adapt hyperparameters to the scenario	Posterior over scenario family from observable signals

The daily cycle follows the two-phase protocol: in the ask phase the policy observes state, runs Layers 1–4 and returns asks; in the match phase it re-observes (so new answers are used), recomputes feasibility, and runs Layers 1–3 to return pairs. Memory is serialised in the response and returned with the next request.

5 Layer 0 — Validity and safety shell

Because validity is binary at the submission level, the shell is built and tested before any model.

- **Independent re-validation.** Before emitting a batch, every pair is checked against all conditions in Section 7 of the specification: distinct available adults in the same pool, both-direction age and gender eligibility, relationship structure, smoking and children constraints, mutual zone acceptance,

schedule overlap, not previously introduced, no outstanding introduction, and no person repeated in the batch. Offending pairs are dropped, not repaired.

- **Exception fallback.** The whole policy is wrapped so that any error results in `{"asks": [], "pairs": []}` (an always-valid response) with diagnostics on stderr only.
- **Time guard.** Each phase tracks elapsed wall-clock time from process start. If a soft deadline (about 6 s of the 10 s limit) is reached, expensive steps (LP futures, VOI) are skipped and a precomputed greedy-on-posterior-mean batch is used.
- **Budget guard.** Asks are trimmed so their total cost never exceeds 12 units; no member is asked a bundle that is already resolved.
- **Memory guard.** Carried memory is pruned (oldest feedback summaries first) to stay under roughly 800 KB, well inside the 1 MiB limit.
- **Determinism.** All randomness is seeded from the declared inference seed, the episode day and the phase, so runs are reproducible.
- **Fuzz testing.** Empty population, empty feasible graph, all-missing profiles, everyone unavailable, every pair already introduced, and maximum-size requests, run both locally and inside the container.

6 Layer 1 — Reciprocal feasibility

Feasibility is a hard gate, implemented with the supplied eligibility function and mirrored in the shell. Each candidate pair is assigned one of three states:

State	Meaning	Use
Feasible	All hard rules satisfied in both directions with current observations	Eligible edge for allocation
Infeasible	At least one known hard failure	Discarded; a known failure takes precedence over missing data
Blocked	No known failure, but a required hard answer is not yet observed	Never matched; feeds the clarification layer

Principles we follow, taken from the data contract:

- A null value is unknown. It never means zero, rejection or lack of interest.
- Field status distinguishes *observed*, *not yet asked* and *declined*. A declined hard answer keeps the pair blocked; we do not infer it.
- Age and gender never imply a preference; only `who_to_meet` defines who a person wants to meet.
- Fictional zones are allocation constraints. Relocation interest does not override currently acceptable zones.
- The feasible graph is general (not bipartite), since eligibility comes from stated preferences rather than assumed categories.

7 Layer 2 — Hierarchical Bayesian outcome model

7.1 Staged factorisation

For a pair (i, j) assigned on day t, MSMI requires mutual acceptance, a date within 30 days, and both people saying Yes within 3 days of the date. By the chain rule (no independence assumption):

$$\mathrm{P}(\mathrm{MSMI}) = \mathrm{P}(\text{mutual accept}) \times \mathrm{P}(\text{date} \le \text{t+30} \mid \text{mutual}) \times \mathrm{P}(\text{both Yes} \le \text{date+3} \mid \text{date})$$

Stage	Target	Model
S1	Each person records Yes by the 7-day deadline; joint mutual acceptance	Bivariate probit with directional linear predictors and correlation ρ (Section 7.3)
S1r	Whether a person responds at all by the deadline	Discrete-time hazard with person responsiveness effect
S2	Date happens within 30 days of assignment, given mutual acceptance	Bayesian logistic regression
S3	Both report Yes within 3 days of the date, given a date	Bivariate probit (or logistic on the joint event)

7.2 Linear predictors and hierarchy

For the direction i → j:

$$\eta(i \to j) = \beta^{T}\, x(i \to j) + u_i + v_j + \delta_e$$

- $x(i \to j)$: directional pair features (Section 7.4).
- $u_i \sim N(0, \tau_u^2)$: person i's general willingness to say Yes (selectivity).
- $v_j \sim N(0, \tau_v^2)$: how generally appealing j is in this synthetic world, learned only from observed responses.
- δ_e: an episode-level offset capturing shifted response conditions (Section 7.6).

Global coefficients β have weakly informative priors $N(0, 1)$ on standardised features; variance components have half-normal priors. Stages S2 and S3 have analogous predictors with their own coefficients and episode offsets.

7.3 Correlated reciprocal acceptance

The specification warns that multiplying directional probabilities assumes independence. We model the two responses with a latent bivariate normal:

$$(z_{ij}, z_{ji}) \sim N((\eta(i \to j), \eta(j \to i)), [[1, \rho], [\rho, 1]]), \quad \text{Yes} \Leftrightarrow z > 0$$

so P(mutual) = $\Phi_2(\eta(i \to j), \eta(j \to i); \rho)$. ρ is estimated offline; if it is near zero, the model reduces to the product form and we report that finding.

7.4 Features

Group	Examples	Notes
Preference margins	Distance of each age from the other's acceptable range edges	Hard eligibility already enforced; margins capture softness near boundaries
Practical fit	Number of overlapping schedule slots, available time (min of both)	Drives date probability (S2)
Soft compatibility	Relationship goal match, pace difference, lifestyle similarity, conversation preference match	Drives S1 and S3
Missingness	One-hot status per soft field: observed / not asked / declined, for each side	Never imputed as a value
History	Shrunken past Yes-rate and response rate, number of prior introductions, days since arrival	Shrinkage via the hierarchy, not raw ratios
Conversation	Template-derived engagement and tone indicators	Treated as an ablation; templates are authored, not realistic speech

7.5 Offline estimation

- Training data comes from simulator rollouts from day zero on the six training pools, using a mixture of logging policies (random-feasible, greedy, and successive BRVM versions) to widen coverage of the pair space.
- Only **pre-action observations** are used: each example records the features observable on the day of assignment, never the final profile.
- Fitting: full Bayesian inference [5] (NUTS or variational) offline; the shipped artefact is a Laplace approximation: posterior means and covariance matrices of global coefficients, plus variance components and ρ. This is a few kilobytes of NumPy arrays.
- Validation pools tune priors and feature sets; development-test pools are touched only for final reporting.

7.6 Online updating within an episode

- **Person effects** (u_i, v_j, responsiveness) are updated after each observed response with a one-dimensional Laplace update holding global coefficients fixed. This is closed-form and cheap, and state per person is two numbers (mean, variance).
- **Episode offsets** δ_e for each stage are updated the same way from all observed outcomes in the episode, letting the policy notice that this world responds or dates at different rates.
- **Censoring.** An introduction contributes to a stage's likelihood only once that stage's observation window has closed. Pending responses contribute a survival term ("no response yet after k days") to the hazard model, not a No. A pair that never dates contributes nothing to S3.

7.7 Optional probability estimates

If reported, we will define each target explicitly with its observation window, and show calibration on held-out pools using matured outcomes only:

Target	Definition	Window
P(i → j)	i records Yes to the introduction	Assignment day t to t+7
P(j → i)	j records Yes to the introduction	t to t+7
P(mutual)	Both record Yes	t to t+7
P(MSMI)	Date within 30 days and both Yes within 3 days of the date	t to at most t+33

A missing response means the specified event did not occur; it is never treated as evidence of dislike.

8 Layer 3 — Allocation with reservation values

8.1 Edge weights

For each feasible pair:

$$w(i,j) = \tilde{p}(i,j) - \kappa\cdot(\lambda_i + \lambda_j) + \varepsilon\cdot(s_i + s_j)$$

- $\tilde{p}(i,j)$: a Thompson sample of P(MSMI) (Section 8.2).
- λ_i: person i's reservation value, the expected MSMI they could contribute if kept free (Section 8.3); κ is a tuned scale.
- s_i: a scarcity score (Section 8.5), with small tuned weight ε.

Only edges with $w > 0$ are kept. If none remain, the policy introduces nobody, which is valid and sometimes optimal.

8.2 Thompson sampling

Following Thompson sampling [2, 3], each day we draw one sample of global coefficients from their Laplace posterior (covariance scaled by a temperature T), and one sample of each relevant person effect, then compute $\tilde{p}$ for every feasible edge. Sampling happens *only over feasible edges*, so exploration can change which valid pair is chosen but can never propose an invalid one. T is tuned and decays toward 0 near the end of the horizon, where exploration has no future payoff.

8.3 Reservation values from LP dual prices

A person's reservation value should reflect how much better their future options are likely to be. We estimate it with a sample-average fluid approximation, in the spirit of dynamic matching for kidney exchange [6, 7]:

1. Sample K future scenarios ($K \approx 10$–20) over a rolling window W (≈ 14 days, truncated at day 59). Future arrivals are drawn from an arrival-rate model fitted on training pools; their profiles are resampled from training-pool members.
2. Build a graph containing current available people plus sampled arrivals. Edge values are posterior-mean P(MSMI), discounted for expected lock-up time.
3. Solve the LP relaxation $\max \sum w\,x$ subject to each person being used at most once ($\sum_j x_{ij} \le 1$), $0 \le x \le 1$, using HiGHS [10] through SciPy.
4. Take the dual price y_i of each current person's capacity constraint, and average across the K scenarios: $\lambda_i = \text{mean_k}\ y_i^{(k)}$.

Interpretation: λ_i is the marginal system value of keeping i free. A person whom many good future partners would want gets a high λ and is not spent on a mediocre pair. A person with few options gets a low λ and is matched readily. As the window shrinks toward day 59, duals fall naturally, producing **endgame aggressiveness** without a hand-coded rule (we also tune an explicit taper as a safeguard).

The LP has at most a few hundred nodes, so it solves in milliseconds. The relaxation ignores odd-cycle constraints, which is acceptable because we only use its duals as prices, not its solution.

8.4 Lock-up and non-responders

An introduction occupies both people until it resolves (up to 7 days). The expected lock-up length uses each person's posterior responsiveness, so pairing with a likely non-responder carries a higher implied cost. Persistent non-responders are therefore deprioritised automatically rather than by a hard rule.

8.5 Scarcity-aware coverage

Coverage is the first tie-breaker and is lost mostly on hard-to-match people. The scarcity score combines low feasible degree and long waiting time:

$$s_i = 1/(1 + \deg(i)) + \min(\text{wait}_i \,/\, 30,\ 1)$$

Its weight ε is small and capped so it can only break near-ties, never override a large MSMI difference. We report the coverage gained and MSMI cost (H7).

8.6 Matching

We run maximum-weight matching on the general graph of positive-weight edges (Edmonds' Blossom algorithm [1], `networkx.max_weight_matching`). Each person appears at most once by construction. To bound runtime, each node keeps its top-k edges ($k \approx 15$–25) by weight, and the time guard falls back to greedy if needed.

9 Layer 4 — Clarification by value of information

We price each clarification by its expected value of information [4]: the expected improvement in matching value once the answer is known, net of the current value.

9.1 Candidate asks

- **Hard-constraint bundle (3 units):** for a member with any unresolved hard field that is blocking at least one pair.
- **Soft field (1 unit):** for a named soft field on a member who is part of a near-threshold edge.

9.2 Value of an ask

For an ask a about member i, with possible outcomes o (each disclosed answer value, or declined):

$$\mathrm{VOI}(\mathrm{a}) = \mathrm{H}(\mathrm{i},\mathrm{t}) \times [\ \Sigma_{\mathrm{o}}\ \mathrm{P}(\mathrm{o}) \cdot \mathrm{V}(\text{after o}) - \mathrm{V}(\text{now})\ ]$$

- `P(o)` comes from a prior-predictive model of the field given i's observed fields, fitted on training pools, including a learned decline probability.
- `V(·)` is the value of i's best available edge (positive w). For speed we approximate it locally as the best newly unlocked or changed edge incident to i; for the top candidates we re-solve the matching exactly when time allows.
- `H(i,t)` is a time-awareness multiplier: the fraction of remaining useful horizon for i (expected days available before day 59), so asking a newcomer early is worth more than asking on day 57.

9.3 Budget allocation

Asks are selected by an exact 0/1 knapsack over items of cost 1 and 3 with capacity 12 (trivial dynamic programming). Asks with non-positive VOI are never made, which keeps clarification cost low (the third tie-breaker). Members who have previously declined a field are not re-asked that field. If budget goes unused on a day, VOI is not inflated to spend it.

9.4 Why this matters most in some families

In cold-start and sparse-geography worlds, many edges are blocked rather than infeasible. Unblocking the right three or four members per day is likely the difference between serving and not serving them, which directly affects both MSMI and coverage.

10 Layer 5 — Scenario-aware meta-policy

The six families (standard, sparse geography, cold start, delayed dates, shifted outcome weights, changing response conditions) probably favour different settings. We therefore:

1. Tune a hyperparameter set θ_f per family on the public variants (Section 12.4).
2. Compute observable signals each day: cumulative arrivals and arrival rate, feasible-graph density, zone diversity, fraction of missing fields, empirical response rate and delay, empirical date lag.
3. Use a small multinomial logistic classifier, trained on public-variant rollouts, to produce a posterior π over families.
4. Blend continuous hyperparameters as `θ = Σ_f π_f θ_f`, starting from the standard-family default on day 0 and committing as evidence accumulates.

Only observable state and policy-owned memory are used, consistent with Section 11 of the specification. If the classifier is uncertain, the blend stays near the robust default.

11 Learning from selective, delayed feedback

11.1 Cold start

New members have no history; their person effects equal the population prior, and their predictions rely on observed profile features. Thompson sampling gives them a fair chance of being explored. Unresolved hard fields make them prime clarification candidates on arrival.

11.2 Selective labels and the policy-improvement loop

The simulator only reveals outcomes for introduced pairs, and no logging propensities are supplied, so we do **not** claim unbiased inverse-propensity evaluation. Instead:

- Training data mixes several logging policies, including random-feasible, so the model sees pairs a tuned policy would avoid.
- We iterate: BRVM v_k generates rollouts → retrain the outcome model → BRVM v_(k+1). We report the score per iteration and stop when gains plateau.
- All policy comparisons are made by **controlled simulator episodes** with identical seeds, never by offline re-weighting of logs.

11.3 Temporal integrity

- Features are snapshotted at the time of each action from rollout logs; the day-30 static snapshots are used for exploration only, never to label earlier decisions.
- Right-censored introductions are excluded from stage labels until their windows close; mature outcomes are evaluated after the 40-day follow-up.

11.4 Separating sources of uncertainty

Source	Representation in BRVM
Model (epistemic) uncertainty	Posterior covariance of coefficients and person effects; drives Thompson sampling
Missing data	Explicit status features; blocked edges; VOI over the distribution of the unknown answer
Stochastic outcomes	Bernoulli likelihoods at each stage; no attempt to predict which coin flip lands

12 Baselines and experimental plan

12.1 Data splits

We use the manifest's split: six training pools, two validation pools, two development-test pools. Populations are never mixed across splits. Training seeds and the inference seed are declared in the repository.

12.2 Baselines

- **Greedy** (supplied), **no-clarification** (supplied) and **random-feasible** (supplied), run with identical seeds and scenario variants through the command-line evaluator.
- **BRVM-lite**: global matching on posterior means with no reservation values or VOI; separates the effect of the model from the effect of the decision machinery.

12.3 Ablations

ID	Change	Hypothesis tested	Expected direction
A1	Blossom matching → greedy ranking	H1	MSMI decreases
A2	$\lambda = 0$ (never wait)	H2	MSMI decreases, most in dense pools
A3	LP-dual λ → heuristic λ (degree-based)	H2	Small decrease
A4	Thompson → posterior mean	H3	Decrease in cold-start / shifted families
A5	VOI asks → random asks (same budget)	H4	Lower MSMI per unit
A6	No asks	H4	Decrease, largest in cold start
A7	No online updates (person or episode)	H5	Decrease under shift
A8	Correlated → independent acceptance	H6	Small or no change; reported either way
A9	Meta-policy → single parameter set	H5	Decrease in non-standard families
A10	Remove conversation features	—	Likely negligible; reported
A11	Remove scarcity bonus	H7	Lower coverage, similar MSMI

12.4 Hyperparameter tuning

Knobs: κ (reservation scale), ε (scarcity weight), T (Thompson temperature) and its decay, endgame taper start day, VOI threshold, top-k edge pruning, and window W. Tuned with Bayesian optimisation (Optuna TPE [8]) on public scenario variants using training and validation pools only. The objective is exactly the ranking statistic: the equal-weighted mean of family means, using common random numbers across candidates.

12.5 Metrics reported

Per family and overall, with per-seed results in machine-readable form:

- Assignments, mutual acceptances, dates, MSMI count and MSMI per 100 arrived members
- Coverage (distinct arrived members introduced ÷ N)
- Clarification cost and asks by type; answers disclosed vs declined
- Missing feedback (non-responses) counts
- Time to first introduction (Kaplan–Meier with unserved members censored) and unserved count
- Inference time per invocation (median, p99, max) inside the container

12.6 Statistical protocol

- **Paired comparisons.** Every policy runs on identical seeds; we report paired differences against each baseline.
- **Uncertainty.** 95% bootstrap confidence intervals over episodes within families, and of the equal-weighted overall score.
- **Small counts.** MSMI counts per episode are small and can be zero; we never treat equal zero counts as evidence of equivalence and report intermediate stages (acceptances, dates) alongside.
- **Calibration** (optional estimates): reliability diagrams and expected calibration error per stage on development-test pools, matured outcomes only.
- **Model checks.** Posterior predictive checks comparing simulated response, date and second-meeting rates with those observed in held-out rollouts.

13 Expected failure cases and mitigations

Failure case	Why it hurts	Mitigation	How we will measure it
Sparse supply / sparse geography	Few feasible edges; waiting rarely pays	Duals fall automatically; VOI targets blocked edges; scarcity bonus	Family-level MSMI and coverage vs greedy
Withheld (declined) answers	Hard-blocked pairs stay blocked permanently	Never infer; avoid re-asking; redirect budget to other members	Share of members permanently blocked; unserved analysis
Delayed feedback / delayed dates	Learning signal arrives late; censoring risk	Censored likelihoods; episode offsets on S2; endgame relies on prior	Delayed-dates family results; calibration of S2
Competition for one candidate	Popular person wasted on a mediocre pair	High dual price λ for that person; global matching	Case studies of contested nodes
Non-responders	Partners locked for 7 days	Responsiveness effect raises lock-up cost	Non-response rate among our introductions
Cold start	No history; uncertain estimates	Hierarchical priors; Thompson exploration; early clarification	Cold-start family; MSMI vs days since arrival
Distribution shift	Offline model miscalibrated in shifted worlds	Online offsets; meta-policy; conservative defaults	Shifted and changing families; offsets over time
Unlucky Thompson draws	Occasional poor pair choices	Temperature cap and decay; samples only over feasible edges	A4 ablation variance across seeds
Runtime spikes	Timeout invalidates the episode	Time guard, edge pruning, greedy fallback	p99 and max inference time in container
Model misspecification	Systematically wrong pair values	Posterior predictive checks; BRVM-lite comparison	Gap between predicted and observed stage rates

The final report will include three to four concrete episodes where BRVM underperformed a baseline, with an explanation of each.

14 Execution budget and engineering

Constraint (spec)	Plan
10 s wall-clock per invocation, including startup	Target < 3 s median; soft deadline at about 6 s triggers greedy fallback
2 CPU cores, 1 GiB memory, CPU only	NumPy, SciPy (HiGHS) and NetworkX only at inference; no PyMC/JAX in the image
1 MiB request, response and memory	Per-person state is a few floats; pruning keeps memory < 800 KB
No persistent filesystem, no network	All model artefacts baked into the image; state only in carried memory
≤ 2 GiB inference assets	Model artefacts are kilobytes; slim Python base image
Docker image from committed Dockerfile; pinned dependencies with licences	Pinned requirements file, licences listed in the repository

Estimated per-invocation costs for a 200-person pool (to be measured): feature construction and feasibility for about 20,000 pairs, vectorised, well under a second; K LP solves of a few hundred variables, under a second in total; pruned Blossom matching, around a second; VOI with local approximation, under a second. Training and tuning run offline on separate compute, which will be documented.

15 Explainability and auditability

For every proposed pair and every ask, the policy writes a structured record to stderr (stdout stays protocol-only), for example:

```
{"day": 23, "pair": ["m_041", "m_117"],
 "p_msmi_mean": 0.31, "p_msmi_sample": 0.34,
 "lambda": [0.08, 0.05], "weight": 0.21,
 "evidence": ["mutual zones", "goal match", "3 shared slots"],
 "unresolved": ["pace: not asked (m_117)"]}
```

Explanations cite only observed evidence and named unresolved fields. They never invent personal history and never present a latent simulator value as an observation. Logs are aggregated in the report into examples of introductions made, deliberate waits, and asks chosen. An optional dashboard over these logs may be added if time allows; it is not required and never replaces the policy.

16 Implementation plan and timeline

Dates (IST)	Work
6–9 Oct	Research (this document). Kit installed; supplied baselines reproduced; rollout data generator working.
12–13 Oct	Validity shell, feasibility, Blossom matching on a simple Bayesian model, endgame taper. First valid container build pinned as an early submission.
14–15 Oct	Full staged hierarchical model; Thompson sampling; LP-dual reservation values; online updates.
16 Oct	VOI clarification; meta-policy; hyperparameter tuning on public variants.
17 Oct	Full evaluation across all six families, baselines and ablations; container verification with the published commands.
18 Oct	Technical report, attribution, final container check; commit SHA pinned well before 23:59 IST.

16.1 Prioritisation

Tier	Components	Status
1	Validity shell, feasibility, global matching, simple Bayesian model, endgame	Committed; submittable on its own
2	LP-dual reservation values; per-family tuning and meta-policy	Committed; expected largest score gains
3	Time-aware VOI clarification; policy-improvement loop	Committed
4	Correlated acceptance, scarcity bonus, conversation features	Stretch; strong analysis value
5	Audit-log dashboard	Optional polish

A valid build is submitted early in Round 2 and replaced as improvements land, so a valid entry is always pinned.

17 Limitations, ethics and scope

- All people and outcomes are synthetic. Simulator performance is not evidence that BRVM can predict real relationships, and its probabilities are not calibrated for real members.
- Conversation data is authored templates; any signal extracted there tests input handling, not psychological inference.
- Hard constraints are never traded off against predicted value. A person with no feasible candidate may remain unmatched, and we report such cases rather than dropping them from coverage statistics.
- Reservation values can delay introductions for in-demand people; we report waiting-time distributions so this trade-off is visible.
- Real deployment would require validation on authorised data, current constraint checks and human review, as the specification states.
- We will document all external libraries, coding tools and compute used, with licences.

18 Open questions to verify in the kit

These details affect parameter choices but not the overall design; we will confirm them from `docs/POLICY_INTERFACE.md`, `docs/DATA_CONTRACT.md` and the public FAQ before Round 2:

- Whether an early No releases both people before the 7-day deadline (changes the lock-up cost in λ).
- Whether paused or departed members can return within an episode.
- How arrivals are generated in each public scenario variant.
- Exact fields covered by the hard-constraint bundle and their disclosure/decline behaviour.

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Appendix A — Notation

Symbol	Meaning
i, j	Members; (i, j) a candidate pair
t	Simulation day of the decision
N, M	Arrived members by day 59; qualifying MSMI introductions
$\eta(i \to j)$	Linear predictor for i saying Yes to j
β, u_i, v_j, δ_e	Global coefficients; selectivity; appeal; episode offset
ρ	Correlation of reciprocal acceptance latents
$\tilde{p}(i,j)$	Thompson-sampled P(MSMI) for the pair
λ_i	Reservation value (mean LP dual price) of member i
κ, ε, T	Reservation scale, scarcity weight, Thompson temperature
w(i,j)	Opportunity-adjusted edge weight used for matching
H(i,t)	Time-awareness multiplier for VOI
π_f, θ_f	Posterior probability of scenario family f; its tuned hyperparameters

Appendix B — Daily decision pseudocode

```
def ask_phase(request):
    mem   = load_memory(request)              # person posteriors, offsets, pi_f
    state = request.state
    update_posteriors(mem, state.feedback)    # censored, conjugate/Laplace
    theta = blend_hyperparams(mem.pi_f)
    F, Blocked = feasibility(state)           # supplied eligibility function
    lam   = lp_dual_reservations(state, F, theta)   # skipped on time guard
    asks  = knapsack(voi_candidates(Blocked, F, lam, theta), budget=12)
    return validate_asks(asks), mem

def match_phase(request):
    mem, state = load_memory(request), request.state  # includes new answers
    F, _  = feasibility(state)
    p     = thompson_sample(F, mem, theta)    # feasible edges only
    w     = p - kappa*(lam_i + lam_j) + eps*(s_i + s_j)
    E     = prune_topk({e: w[e] for e in F if w[e] > 0})
    pairs = blossom_max_weight_matching(E)
    pairs = revalidate(pairs, state)          # drop anything invalid
    log_explanations(pairs)                   # stderr only
    return pairs, mem

# Any exception or time-guard breach in either phase returns an empty,
# valid response (or the precomputed greedy batch in the match phase).
```